~~Product Name:~~ ~~LEUCOVORIN/CALCIUM FOLINATE 10mg/ml~~	~~Territory:~~ ~~MY~~	~~Colour:~~ ● ~~BLACK~~ ● ~~DIE CUT~~	~~1. Draft 07.11.2018 08.00 Uhr~~ ~~1. Correction 13.11.2018 10.00 Uhr~~ ~~2. Correction 14.11.2018 10.20 Uhr~~ ~~3. Correction 15.11.2018 08.22 Uhr~~		~~Variable Data:~~
~~Type of Packaging:~~ ~~PIL~~	~~Dosage:~~ ~~10mg/ml~~				
~~Material number:~~ ~~M080357/00 MY~~	~~2-D Matrix Code~~ ~~M080357/00 MY~~				
~~Pharma Code (Laetus)~~ ~~x~~	~~EAN Code:~~				
~~Dimension:~~ ~~180 x 588 mm~~	~~Font: 8 Pt.~~ ~~Smallest Size: Arial~~		Operator: **Christian Nagy** +43(0) 34 52 72266-22		

For the use only a Registered Medical Practitioner or a Hospital or a Laboralory

Leucovorin Kabi Injection USP 10mg/ml

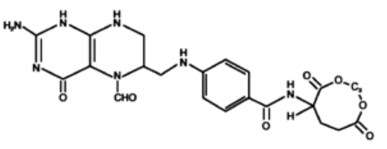
DESCRIPTION:
Leucovorin Kabi Injection USP 10mg/ml is sterile, preservative-free solution indicated for intramuscular(IM) or intravenous(IV) administration in single-dose vials. Leucovorin Calcium Injection is clear,yellowish solution. After dilution, Leucovorin Calcium Injection is clear colorless to yellowish solution.

COMPOSITION

Each 5 ml vial contains:	
Leucovorin Calcium USP	
equivalent to Leucovorin	50mg
Sodium Chloride USP	0.8%w/v
Methylparaben USP	0.08%w/v
Propylparaben USP	0.02%w/v
Water for injection USP	q.s.to 5 ml

CHEMICAl STRUCTURE
Leucovorin Calcium is chemically known as calcium N-[4•(2-amino-5-fonnyl-1,4,5,6,7,8 hexahydro4- oxo-6ptredinyl) methylamino-benzoyi)•L-glutamate.

The molecular fomula is $C_{20}H_{2}PaNp_{7}$ and having a molecular weight of 511.51. Leucovorin Calcium is a yellow-white or yellow powder, very solubie in water, practically insoluble in alcohol.



PHARMACOLOGY MECHANISM OF ACTION
Leucovorin calcium is a reduced form of folic acid. As such it competes effectively with Methotrexate for transport into mammalian cells and presumably human target tissues such as bone marrow and gastrointestinal epitheliurn. In the cancer patient, Leucovorin calcium is used to inhibit the effects of methotrexate when given in high, otherwise lethal doses.
Methotrexate by binding to dihydrofolic acid reductase inhibits the production of reduced folates. Subsequently, thymidine synthesis is halted and DNA, RNA and protein synthesis may be interrupted. The affinity of dihydrofolate reductase for methotrexate is far greater than its affinity for folic acid or dihyrofolic acid and, therefore, even very large doses of folic acid given simultaneously will not reverse the effects of methotrexate. Reduction by dihydrofolate reductase is not required by leucovorin and hence it is metabolized to tetrahydrofolate there by passing enzymatic block induced by methotrexate. Leucovorin thus prevents and "rescue" cells from adverse effects of methotrexate.

PHARMACOKINETICS
After intravenous administration, leucovorin is distributed to all tissues and crosses the blood brain barrier. /-leucovorin (l-5-formyltetrahydrofolate) is rapidly metabolized to 1-5- methyl tetrahydrofol ate.This is further metabolized to 5,10 methylenetetrahydrofol ate which is converted to 5-methyltetrahydrofolate by an irreversible enzyme catalyzed reduction using the co-factors FADH2 and NADPH. The mean peak serum total reduced folates was 1259ng/mL and the mean time to peak was 10 minutes. The initial rise in total reduced folates was primarily due to the parent compound 5-forml-tetrahydrofolate which rose to 1206ng/mL at 10 minutes. The was followed by a sharp drop in parent compound and coincided with the appearance of the active metabolite 5-methyl-tetrahydrofolate which became the predominant circulating form of the drug. The mean peak of 5- methyl- tetrahydrofolate was 258 ng/mL and occurred at 1.3 hrs. The terminal half-life for total reduced folates was 6.2 hours. After intramuscular injection, the mean peak of serum total reduced folates was 436 ng/mL and occurred at 52 minutes. Similar to IV administration, the initial sharp rise was due to the parent compound. The level of the metabolite 5-methyl- tetrahydrofolate increased subsequently over time until at 1.5 hrs it represented 50% of the circulating total folates. The mean peak of 5-methyl-tetrahydrofolate was 226 ng/mL at 2.8 hrs. the terminal half-life of total reduced folate was 6.2 hrs. The metabolites are rapidly eliminated in urine.

DRUG INTERACTIONS
- Leucovorin increases cytotoxic and toxic effects of fluorouracil by stabilizing the bond of thymidilate synthetase
- Toxicity of methotrexate is decreased by leucovorin as it rescues normal cells from toxic effect of methotrexate
- Concomitant therapy with folic acid antagonist (e.g Trimetoprim, co-trimaxazole) can reduce the efficacy of folic acid antagonist.

CLINICAL STUDIES
A study was conducted to determine the effectiveness of the 5-FU+leucovorin combination as adjuvant therapy following surgery as compared to single agent 5-FU in treatment of patients with advanced colorectal cancer: 32 patients who were receiving 5-FU(375 mg/m2) oral intake with low dose leucovorin (20 mg/m2) intravenously for 5 days regimen were associated with an improved survival and less recurrence in liver and lung compared to 61 patients treated with single 5-FU oralintake($p<0.05$).
High dose leucovorin, administered early after the diagnosis of methotrexate-induced nephrotoxicity, was effective in reducing toxicity in all of the 13 patients without the use of extracorporeal removal of methotrexate (ie, hemodialysis, hemofiltration). No patient died and most patients suffered side effects (mucositis, nausea, vomiting, diarrhea), presumably form methotrexate treatment.

INDICATIONS Leucovorin Kabi Injection is indicated:

(a) to diminish the toxicity and counteract the effects of inadvertently administered over dosage of folic acid antagonists,
(b) as a rescue after high dose methotrexate therapy.
(c) in the treatment of megaloblastic anaemia due to sprue, nutritional deficiency, pregnancy and infancy when oral therapy is not possible and,
(d) for use in combination with fluorouracil to prolong survival in the palliative treatment of patients with advanced colorectal cancer.

CONTRAINDICATIONS
1. It is contraindicated in patients who have known hypersensitivity to Leucovorin
2. Megaloblastic anaemia due to cyanocobalamine deficiency
3. Pernicious anaemia due to cyanocobalamine deficiency

ADVERSE REACTIONS
Stomatitis, diarrhea, enhanced myelosuppression, skin rash. hives, pruritis have been reported following administration of Leucovorin Calcium.

PRECAUTIONS ANDWARNINGS
- In the treatment of accidental overdosages of folic acid antagonists, leucovorin should be administered as promptly as possible. As the time interval between antifolate administration (e.g. Methotrexate) and leucovorin rescue increases, leucovorin's effectiveness in counteracting toxicity decreases. Monitoring of the serum methotrexate concentration is essential in determining the optimal dose and duration of treatment with leucovorin. Delayed methotrexate excretion may be caused by a third space fluid accumulation (i.e. ascites, pleural effusion), renal insufficiency or inadequate hydration. Under such circumstances, higher doses of leucovorin or prolonged administration may be indicated
- Because of the calcium content of the leucovorin solution, not more than 160 mg of leucovorin should be injected intravenously per minute
- Leucovorin/5-fluorouracil combination therapy should be administered under the supervision of a physician experienced in the use of antimetabolite cancer chemotherapy. Particular care should be taken in the treatment of elderly or debilitated patients, as these patients may be at increased risk of severe toxicity. Patients treated with Fu and leucovorin combination should have a CBC differential and platelets prior to each treatment and has to be repeated weekly during the first two courses and thereafter once each cycle
- Folic acid in large amounts may counteract the antiepileptic effect of phenobarbital, phenytoin and primidone and increase the frequency of seizures in susceptible children
- Leucovorin is harmful or fatal if given intrathecally and hence it is not recommended
- In cases of pernicious anaemia and megaloblastic anaemia secondary to Vit b12 deficiency leucovorin can enhance the hematologic effects of Vit b12 deficiency while allowing the neurological complications to continue
- Serum crealinine and Methotrexate levels should be determined at least once daily.
- Electrolytes and liver function tests should be performed prior to first three cycles of each treatment and then prior to each other cycle.

Pregnancy and lactation
Pregnancy Category C: Adequate animal reproduction studies have not been conducted with leucovorin. It is also not known whether leucovorin can cause foetal harm when administered to a pregnant woman or can affect reproduction capacity. Leucovorin should be given to a pregnant woman only if clearly needed.
Nursing Mothers: It is not known whether this drug is excreted in human milk, therefore, caution should be exercised when leucovorin is administered to a nursing mother.

<u>Impaired Methotrexate Elimination or Inadvertent</u> Overdosage: Leucovorin rescue should begin as ~~soon as pos~~sible alter an inadvertent overdosage and within 24 hours of Methotrexate administration when there is delayed excretion. Leucovorin 10 mg/m2 should be administered I.V. or I.M. every 6 hours until the serum methotrexate level is less than 10-8 M. Serum creatinine and methotrexate levels should be determined at 24-hour intervals. If the 24 hour serum creatinine has increased 50% over baseline or If the 24-hour methotrexate level is greater than 5 X 10-6 or the 48-hour level is greater than 9 x 10-7M, the dose of Leucovorin should be increased to 100 mg/m2 I.V. every 3 hours until the Methotrexate level is less than 10-8 M Hydration (3L/d) and urinary alkalinization with sodium bicarbonate solution should be employed concomitantly. The bicarbonate dose should be adjusted to maintain the urine pH at 7.0 or greater.

<u>Leucovorin Rescue After High-dose Methotrexate Therapy</u>: The recommendations for Leucovorin rescue are based on a methotrexate dose of 12 to 15 gm/m2 administered by intravenous infusion over 4 hours. Leucovorin rescue at a dose of 15 mg (approximately 10 mg/m2) every 6 hours for 10 doses starts 24 hours alter the beginning of the Methotrexate infusion. Serum creatinine & methotrexate levels should be determined at least once daily. Leucovorin administration, hydration, and urinary alkalinization (pH of 7.0 or greater) should be continued until the methotrexate level is below 5 x 10-8 M (0.05 micromolar).

<u>Megaloblastic Anaemia due to Folic Acid Deficiency:</u>
Upto 1 mg daily. There is no evidence that doses greater than 1 mg/day have greater efficacy than those of 1 mg, additionally loss of folate in urine becomes roughly logarithmic as the amount administered exceeds 1 mg.

Advanced Colorectal Cancer
Either of the following two regimens is recommended:
1. Leucovorin is administered at 200 mg/m2 by slow intravenous injection over a minimum 3 minutes, followed by 5 Fluorouracil at 370 mg/m2 by intravenous injection.
2. Leucovorin is administered at 20 mg/m2 by intravenous injection followed by 5- fluorouracil at 425 mg/m^2 by intravenous injection.

Treatment is repeated daily for five days. This five days treatment course may be repeated at 4-week (28 days) intervals for two courses and then repeated at 4 to 5 week (28-35 days) intervals provided that the patient has completely recovered form toxic effects of the prior treatment course.

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In subsequent treatment courses, the dosage of 5 fluorouracil should be adjusted based on patient intolerance of the prior treatment course. The daily dosage of 5-fluorouracil should be reduced by 20% for patients who experienced moderate haematologic or gastrointestinal toxicity in the prior treatment course and by 30% for patients who experienced severe toxicity. For patients who experienced no toxicity in prior treatment course, 5 Fluorouracil dosage may be increased by 10%. Leucovorin dosage are not adjusted for toxicity.

ROUTE OF ADMINISTRATION:
Intravenously(IV), Intramuscularly (IM)

OVERDOSAGE
Excessive amounts of leucovorin may nullify the chemotherapeutic effect of folic acid antagonists.

INCOMPATIBILITIES:
Leucovorin should not be mixed in the same infusion as 5-fluorouracil, since this may lead to the formation of a precipitate.

SHELF LIFE:
18 months.

After first opening:
For single dose use only. Discard any unused solution immediately after initial use.

After dilution:
When diluted according to directions with 0.9% sodium chloride injection or 5% glucose injection, chemical and physical in-use stability has been demonstrated when protected from light. Chemical and physical in-use stability after dilution to 1.5 mg/ml with either 0.9% sodium chloride injection or 5% glucose injection was demonstrated for up to 24 hours, at both room temperature (25°C) and 2°C- 8°C, when protected from light.
Chemical and physical in-use stability after dilution to 0.2 mg/ml with 0.9% sodium chloride injection was demonstrated for up to 24 hours at 2°C - 8°C, when protected from light.
From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2°C -8°C, unless dilution has taken place in controlled and validated aseptic conditions.

STORAGE:
Store at 2-8°C. Do not freeze. Protect from light.

PRESENTATION:
Leucovorin Kabi Injection USP 10mg/ml are available as 5 ml, 10 ml, 20 ml, 35 ml, 50 ml and 100 ml vial containing 50 mg,
100 mg, 200 mg, 350 mg, 500 mg and 1000 mg of leucovorin respectively.
Pack sizes:
10 x 5 ml
10 x 10 ml
10 x 20 ml
10 x 35 ml
10 x 50 ml
10 x 100 ml
Not all pack sizes may be marketed.

MANUFACTURER INFORMATION:
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